

Experiments

1. Lists, Links and Images

a. Write a HTML program, to explain the working of lists.

Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Lists Example</title>
</head>
<body>

<h1>HTML Lists Example</h1>

<h2>Ordered List</h2>
<ol>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ol>

<h2>Unordered List</h2>
<ul>
  <li>Apple</li>
  <li>Banana</li>
  <li>Orange</li>
</ul>

<h2>Nested List (Ordered inside Unordered)</h2>
<ul>
  <li>Fruits
    <ol>
      <li>Apple</li>
      <li>Banana</li>
    </ol>
  </li>
  <li>Vegetables
    <ul>
      <li>Carrot</li>
      <li>Potato</li>
    </ul>
  </li>
</ul>

<h2>Ordered List inside Unordered List</h2>
<ul>
  <li>
```

```

        <ol>
        <li>Red</li>
        <li>Green</li>
        <li>Blue</li>
        </ol>
    </li>
    <li>
        <ol>
        <li>Square</li>
        <li>Circle</li>
        <li>Triangle</li>
        </ol>
    </li>
</ul>

<h2>Definition List</h2>
<dl>
    <dt>HTML</dt>
    <dd>Hypertext Markup Language</dd>
    <dt>CSS</dt>
    <dd>Cascading Style Sheets</dd>
</dl>

</body>
</html>

```

b. Write a HTML program, to explain the working of hyperlinks using <a> tag and href, target Attributes.

```

<!DOCTYPE html>
<html>
<head>
    <title>Hyperlink Example</title>
</head>
<body>

    <h1>HTML Hyperlink Demonstration</h1>

```

```

    <p>This is a basic example of using the <code>&lt;a></code> tag with <code>href</code> and
    <code>target</code> attributes.</p>

```

```

    <!-- Hyperlink that opens in the same tab -->
    <p><a href="https://www.wikipedia.org" target="_self">Open Wikipedia in the same tab
    (target="_self")</a></p>

```

```

    <!-- Hyperlink that opens in a new tab -->
    <p><a href="https://www.google.com" target="_blank">Open Google in a new tab
    (target="_blank")</a></p>

```

```

    <!-- Hyperlink that opens in a new window/tab depending on browser settings -->

```

```
<p><a href="https://www.openai.com" target="_blank">Visit OpenAI website</a></p>
```

```
</body>
```

```
</html>
```

- c. **Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.**

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Profile Images with Links</title>
```

```
</head>
```

```
<body>
```

```
  <h2>Click on the images to view profiles</h2>
```

```
  <!-- Your image linked to your profile -->
```

```
  <a href="https://your-profile-link.com" target="_blank">
```

```
    
```

```
  </a>
```

```
  <!-- Friend's image linked to friend's profile -->
```

```
  <a href="https://friend-profile-link.com" target="_blank">
```

```
    
```

```
  </a>
```

```
</body>
```

```
</html>
```

- d. **Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique**

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Image Gallery with Thumbnails</title>
```

```
</head>
```

```
<body>
```

```
  <h1>My Image Gallery</h1>
```

```
  <p>Click on the thumbnails to view full-size images.</p>
```

```
  <!-- Image 1 -->
```

```
  <a href="images/fullsize1.jpg" target="_blank">
```

```
    
```

```
  </a>
```

```

<!-- Image 2 -->
<a href="images/fullsize2.jpg" target="_blank">
  
</a>

<!-- Image 3 -->
<a href="images/fullsize3.jpg" target="_blank">
  
</a>

<!-- Image 4 -->
<a href="images/fullsize4.jpg" target="_blank">
  
</a>

</body>
</html>

```

2. HTML Tables, Forms and Frames

- a. Write a HTML program, to explain the working of tables. (use tags: <table>, <tr>, <th>, <td> and attributes: border, rowspan, colspan)

```

<!DOCTYPE html>
<html>
<head>
  <title>HTML Table Example</title>
</head>
<body>

  <h2>HTML Table with rowspan and colspan</h2>

  <table border="1">
    <tr>
      <th>Day</th>
      <th>Time</th>
      <th colspan="2">Subject</th>
    </tr>

    <tr>
      <td rowspan="2">Monday</td>
      <td>9:00 - 10:00</td>
      <td>Math</td>
      <td>Physics</td>
    </tr>

    <tr>
      <td>10:00 - 11:00</td>
      <td colspan="2">Chemistry</td>
    </tr>
  </table>

```

```

</tr>

<tr>
  <td>Tuesday</td>
  <td>9:00 - 10:00</td>
  <td>Biology</td>
  <td>English</td>
</tr>
</table>

</body>
</html>

```

- b. Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).**

```

<!DOCTYPE html>
<html>
<head>
  <title>School Timetable</title>
</head>
<body>

  <h2>HTML Table Timetable Example</h2>

  <table border="2" cellspacing="5" cellpadding="10">
    <caption><strong>Class Timetable</strong></caption>

    <tr>
      <th>Day</th>
      <th>9:00 - 10:00</th>
      <th>10:00 - 11:00</th>
      <th>11:00 - 12:00</th>
      <th>12:00 - 1:00</th>
      <th>1:00 - 2:00</th>
    </tr>

    <tr>
      <td>Monday</td>
      <td>Math</td>
      <td>English</td>
      <td>Physics</td>
      <td rowspan="5" style="text-align: center;"><strong>Lunch Break</strong></td>
      <td>Biology</td>
    </tr>

    <tr>
      <td>Tuesday</td>
      <td>History</td>

```

```

        <td>Geography</td>
        <td>Chemistry</td>
        <td>Math</td>
    </tr>

    <tr>
        <td>Wednesday</td>
        <td colspan="3" style="text-align: center;">Lab Session</td>
        <td>English</td>
    </tr>

    <tr>
        <td>Thursday</td>
        <td>Computer</td>
        <td>Math</td>
        <td>Physics</td>
        <td>Chemistry</td>
    </tr>

    <tr>
        <td>Friday</td>
        <td>Biology</td>
        <td>English</td>
        <td>Computer</td>
        <td>Games</td>
    </tr>

</table>

</body>
</html>

```

- c. Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select>&<option> tags, <text area> and two buttons ie: submit and reset. Use tables to provide a better view).

```

<!DOCTYPE html>
<html>
<head>
    <title>Registration Form</title>
</head>
<body>

    <h2>Student Registration Form</h2>

    <form action="#" method="post">
        <table border="1" cellpadding="10" cellspacing="0">
            <tr>
                <td><label for="fname">First Name:</label></td>

```

```

        <td><input type="text" id="fname" name="fname" required></td>
    </tr>

    <tr>
        <td><label for="lname">Last Name:</label></td>
        <td><input type="text" id="lname" name="lname" required></td>
    </tr>

    <tr>
        <td><label for="password">Password:</label></td>
        <td><input type="password" id="password" name="password" required></td>
    </tr>

    <tr>
        <td><label for="age">Age:</label></td>
        <td><input type="number" id="age" name="age" min="1" max="100"></td>
    </tr>

    <tr>
        <td><label for="dob">Date of Birth:</label></td>
        <td><input type="date" id="dob" name="dob"></td>
    </tr>

    <tr>
        <td>Gender:</td>
        <td>
            <input type="radio" id="male" name="gender" value="Male">
            <label for="male">Male</label>
            <input type="radio" id="female" name="gender" value="Female">
            <label for="female">Female</label>
        </td>
    </tr>

    <tr>
        <td>Hobbies:</td>
        <td>
            <input type="checkbox" id="reading" name="hobbies" value="Reading">
            <label for="reading">Reading</label>
            <input type="checkbox" id="sports" name="hobbies" value="Sports">
            <label for="sports">Sports</label>
            <input type="checkbox" id="music" name="hobbies" value="Music">
            <label for="music">Music</label>
        </td>
    </tr>

    <tr>
        <td><label for="country">Country:</label></td>
        <td>
            <select id="country" name="country">
                <option value="">--Select Country--</option>
                <option value="India">India</option>
            </select>
        </td>
    </tr>

```

```

        <option value="USA">USA</option>
        <option value="UK">UK</option>
        <option value="Australia">Australia</option>
    </select>
</td>
</tr>

<tr>
    <td><label for="address">Address:</label></td>
    <td><textarea id="address" name="address" rows="4" cols="30"></textarea></td>
</tr>

<tr>
    <td colspan="2" style="text-align: center;">
        <input type="submit" value="Submit">
        <input type="reset" value="Reset">
    </td>
</tr>
</table>
</form>

</body>
</html>

```

- d. Write a HTML program, to explain the working of frames, such that page is to be divided into 3 parts on either direction. (Note: first frame image, second frame □ paragraph, third frame □ hyperlink. And also make sure of using “no frame” attribute such that frames to be fixed).

main.html (Main file with <frameset>)

```

<!DOCTYPE html>
<html>
<head>
    <title>Frameset Example</title>
</head>
<frameset rows="33%,34%,33%" frameborder="1" noresize>
    <frame src="image.html" name="imageFrame">
    <frame src="paragraph.html" name="paraFrame">
    <frame src="hyperlink.html" name="linkFrame">

<noframes>
    <body>
        <p>Your browser does not support frames.</p>
    </body>
</noframes>
</frameset>
</html>

```

image.html (Displays an image)


```

html
CopyEdit
<!DOCTYPE html>
<html>
<head><title>Image Frame</title></head>
<body>
  <h3>My Picture</h3>
  
</body>
</html>

```

Replace "your-image.jpg" with a valid image file path or online image URL.

paragraph.html (Displays a paragraph)

```

html
CopyEdit
<!DOCTYPE html>
<html>
<head><title>Paragraph Frame</title></head>
<body>
  <h3>About Me</h3>
  <p>I am a student learning HTML. This is a sample paragraph displayed in the second frame to explain the concept of frames.</p>
</body>
</html>

```

hyperlink.html (Displays a hyperlink)

```

html
CopyEdit
<!DOCTYPE html>
<html>
<head><title>Link Frame</title></head>
<body>
  <h3>Useful Link</h3>
  <a href="https://www.wikipedia.org" target="_blank">Visit Wikipedia</a>
</body>
</html>

```

3. HTML 5 and Cascading Style Sheets, Types of CSS

- Write a HTML program, that makes use of <article>, <aside>, <figure>, <figcaption>, <footer>, <header>, <main>, <nav>, <section>, <div>, tags.
- Write a HTML program, to embed audio and video into HTML web page.

- c. Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value).

4. Selector forms

- a. Write a program to apply different types of selector forms
 - i. Simple selector (element, id, class, group, universal)
 - ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
 - iii. Pseudo-class selector
 - iv. Pseudo-element selector
 - v. Attribute selector

5. CSS with Color, Background, Font, Text and CSS Box Model

- a. Write a program to demonstrate the various ways you can reference a color in CSS.
- b. Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.
- c. Write a program using the following terms related to CSS font and text:
 - i. font-size ii. font-weight iii. font-style
 - iv. text-decoration v. text-transformation vi. text-alignment
- d. Write a program, to explain the importance of CSS Box model using
 - i. Content ii. Border iii. Margin iv. padding

6. Applying JavaScript - internal and external, I/O, Type Conversion

- a. Write a program to embed internal and external JavaScript in a web page.

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Embedding Example</title>

  <!-- Internal JavaScript -->
  <script>
    function showInternalMessage() {
      alert("Hello from Internal JavaScript!");
    }
  </script>

  <!-- External JavaScript -->
  <script src="external.js"></script>
</head>
<body>

  <h1>Embedding JavaScript</h1>

  <!-- Button for Internal JS -->
  <button onclick="showInternalMessage()">Click for Internal JS</button>
```

```

    <!-- Button for External JS -->
    <button onclick="showExternalMessage()">Click for External JS</button>

</body>
</html>
external.js :

function showExternalMessage() {
    alert("Hello from External JavaScript!");
}

```

- b. Write a program to explain the different ways for displaying output.

```

<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Output Methods</title>
</head>
<body>

    <h2>JavaScript Output Methods</h2>

    <!-- Method 1: Using innerHTML -->
    <p id="demo"></p>

    <!-- Method 2: Using document.write() -->
    <button onclick="documentWriteExample()">Click to use document.write()</button>

    <!-- Method 3: Using alert box -->
    <button onclick="alertExample()">Click to see alert()</button>

    <!-- Method 4: Using console.log() -->
    <button onclick="consoleLogExample()">Click to log in console</button>

    <script>
        // Method 1: innerHTML
        document.getElementById("demo").innerHTML = "1. Output using innerHTML";

        // Method 2: document.write()
        function documentWriteExample() {
            document.write("2. Output using document.write()");
        }

        // Method 3: alert()
        function alertExample() {

```

```

        alert("3. Output using alert()");
    }

    // Method 4: console.log()
    function consoleLogExample() {
        console.log("4. Output using console.log()");
    }
</script>

</body>
</html>

```

- c. Write a program to explain the different ways for taking input.

```

<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Input Methods</title>
</head>
<body>

    <h2>JavaScript Input Methods</h2>

    <!-- Method 1: Using prompt() -->
    <button onclick="usePrompt()">Take Input using prompt()</button><br><br>

    <!-- Method 2: Using form input (text box) -->
    <form onsubmit="return useFormInput()">
        <label for="name">Enter your name (Form Input): </label>
        <input type="text" id="nameInput" required>
        <input type="submit" value="Submit">
    </form><br>

    <!-- Method 3: Using HTML input + event listener -->
    <label for="age">Enter your age (Event Listener): </label>
    <input type="number" id="ageInput">
    <button onclick="useEventInput()">Submit</button><br><br>

    <p id="output"></p>

    <script>
        // Method 1: prompt()
        function usePrompt() {
            let userInput = prompt("Enter something:");
            document.getElementById("output").innerHTML = "You entered (prompt): " +
userInput;

```

```

    }

    // Method 2: Form Input
    function useFormInput() {
        let name = document.getElementById("nameInput").value;
        document.getElementById("output").innerHTML = "You entered (form input): " +
name;
        return false; // prevent form from refreshing the page
    }

    // Method 3: Input + Event Handler
    function useEventInput() {
        let age = document.getElementById("ageInput").value;
        document.getElementById("output").innerHTML = "You entered (event input): " +
age;
    }
</script>

</body>
</html>

```

- d. Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

```

<!DOCTYPE html>
<html>
<head>
    <title>Voter Eligibility Check</title>
    <style>
        table, th, td {
            border: 1px solid black;
            border-collapse: collapse;
            padding: 8px;
        }
        th {
            background-color: #f2f2f2;
        }
    </style>
</head>
<body>

    <h2>Voter Information</h2>
    <div id="voterInfo"></div>

    <script>
        // Prompt the user for name and age

```

```

let name = prompt("Enter your name:");
let age = prompt("Enter your age:");

// Convert age to a number
age = parseInt(age);

// Determine voter eligibility
let eligibility = (age >= 18) ? "Eligible to Vote" : "Not Eligible to Vote";

// Create table HTML
let table = `
    <table>
        <tr>
            <th>Name</th>
            <th>Age</th>
            <th>Voter Status</th>
        </tr>
        <tr>
            <td>${name}</td>
            <td>${age}</td>
            <td>${eligibility}</td>
        </tr>
    </table>
`;

// Display the table on the page
document.getElementById("voterInfo").innerHTML = table;
</script>

</body>
</html>

```

7. JavaScript Pre-defined and User-defined Objects

- a. Write a program using document object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
    <title>Document Object Example</title>
</head>
<body>

    <h2>Document Object Properties and Methods</h2>

    <button onclick="showDocumentInfo()">Click to Show Document Info</button>

```

```

<div id="output" style="margin-top: 20px; font-family: monospace;"></div>

<script>
  function showDocumentInfo() {
    let info = `
      <strong>Document Title:</strong> ${document.title}<br>
      <strong>Document URL:</strong> ${document.URL}<br>
      <strong>Document Domain:</strong> ${document.domain}<br>
      <strong>Last Modified:</strong> ${document.lastModified}<br>
      <strong>Number of Forms:</strong> ${document.forms.length}<br>
      <strong>Number of Links:</strong> ${document.links.length}<br>
    `;

    // Display info in the div
    document.getElementById("output").innerHTML = info;

    // Modify title using document property
    document.title = "Updated Title with JS";
  }
</script>

</body>
</html>

```

- b. Write a program using window object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
  <title>Window Object Example</title>
</head>
<body>

  <h2>Window Object Properties and Methods</h2>

  <button onclick="showWindowProperties()">Show Window Properties</button>
  <button onclick="openNewWindow()">Open New Window</button>
  <button onclick="showAlert()">Show Alert</button>
  <button onclick="confirmAction()">Show Confirm</button>
  <button onclick="promptUser()">Show Prompt</button>

  <div id="output" style="margin-top: 20px; font-family: monospace;"></div>

  <script>
    // Display basic window properties
    function showWindowProperties() {

```

```

let info = `
  <strong>Window Inner Width:</strong> ${window.innerWidth}px<br>
  <strong>Window Inner Height:</strong> ${window.innerHeight}px<br>
  <strong>Window Outer Width:</strong> ${window.outerWidth}px<br>
  <strong>Window Outer Height:</strong> ${window.outerHeight}px<br>
  <strong>Screen X:</strong> ${window.screenX}px<br>
  <strong>Screen Y:</strong> ${window.screenY}px<br>
  <strong>Location:</strong> ${window.location.href}<br>
`;
document.getElementById("output").innerHTML = info;
}

// Method: Open a new window
function openNewWindow() {
  window.open("https://www.example.com", "_blank", "width=500,height=400");
}

// Method: Show alert
function showAlert() {
  window.alert("This is an alert message!");
}

// Method: Show confirm
function confirmAction() {
  let result = window.confirm("Do you want to continue?");
  document.getElementById("output").innerHTML = "User clicked: " + (result ? "OK" :
"Cancel");
}

// Method: Prompt user for input
function promptUser() {
  let name = window.prompt("What is your name?");
  document.getElementById("output").innerHTML = "Hello, " + name;
}
</script>

</body>
</html>

```

c. Write a program using array object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
  <title>Array Object Example</title>

```



```
</head>
```

```
<body>
```

```
<h2>Array Object Properties and Methods</h2>
```

```
<button onclick="displayArray()">Display Original Array</button>
```

```
<button onclick="sortArray()">Sort Array</button>
```

```
<button onclick="reverseArray()">Reverse Array</button>
```

```
<button onclick="pushElement()">Push Element</button>
```

```
<button onclick="popElement()">Pop Element</button>
```

```
<button onclick="joinArray()">Join Array</button>
```

```
<div id="output" style="margin-top: 20px; font-family: monospace;"></div>
```

```
<script>
```

```
    // Declare an array
```

```
    let fruits = ["Apple", "Mango", "Banana", "Orange"];
```

```
    // Display the original array
```

```
    function displayArray() {
```

```
        document.getElementById("output").innerHTML = `
```

```
            <strong>Array Elements:</strong> ${fruits}<br>
```

```
            <strong>Length:</strong> ${fruits.length}
```

```
        `;
```

```
    }
```

```
    // Sort the array
```

```
    function sortArray() {
```

```
        fruits.sort();
```

```
        document.getElementById("output").innerHTML = "Sorted Array: " + fruits;
```

```
    }
```

```
    // Reverse the array
```

```
    function reverseArray() {
```

```
        fruits.reverse();
```

```
        document.getElementById("output").innerHTML = "Reversed Array: " + fruits;
```

```
    }
```

```
    // Add an element using push()
```

```
    function pushElement() {
```

```
        fruits.push("Pineapple");
```

```
        document.getElementById("output").innerHTML = "After push(): " + fruits;
```

```
    }
```

```
    // Remove the last element using pop()
```

```
    function popElement() {
```

```

        let popped = fruits.pop();
        document.getElementById("output").innerHTML = `After pop(): ${fruits} <br> Popped
Element: ${popped}`;
    }

    // Join array elements using join()
    function joinArray() {
        let joined = fruits.join(" - ");
        document.getElementById("output").innerHTML = "Joined Array: " + joined;
    }
</script>

</body>
</html>

```

d. Write a program using math object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
    <title>Math Object Example</title>
</head>
<body>

    <h2>Math Object Properties and Methods</h2>

    <button onclick="showMathProperties()">Show Math Properties</button>
    <button onclick="useMathMethods()">Use Math Methods</button>
    <div id="output" style="margin-top: 20px; font-family: monospace;"></div>

    <script>
        function showMathProperties() {
            const properties = `
                <strong>Math.PI:</strong> ${Math.PI}<br>
                <strong>Math.E:</strong> ${Math.E}<br>
                <strong>Math.LN2:</strong> ${Math.LN2}<br>
                <strong>Math.SQRT2:</strong> ${Math.SQRT2}<br>
                <strong>Math.LOG10E:</strong> ${Math.LOG10E}<br>
            `;
            document.getElementById("output").innerHTML = properties;
        }

        function useMathMethods() {
            let a = 7.8;
            let b = -5;

            const methods = `

```

```

        <strong>Math.round($ {a}):</strong> ${Math.round(a)}<br>
        <strong>Math.floor($ {a}):</strong> ${Math.floor(a)}<br>
        <strong>Math.ceil($ {a}):</strong> ${Math.ceil(a)}<br>
        <strong>Math.abs($ {b}):</strong> ${Math.abs(b)}<br>
        <strong>Math.sqrt(64):</strong> ${Math.sqrt(64)}<br>
        <strong>Math.pow(2, 4):</strong> ${Math.pow(2, 4)}<br>
        <strong>Math.max(10, 50, 100):</strong> ${Math.max(10, 50, 100)}<br>
        <strong>Math.min(10, 50, 100):</strong> ${Math.min(10, 50, 100)}<br>
        <strong>Math.random():</strong> ${Math.random()} (between 0 and 1)<br>
    `;
    document.getElementById("output").innerHTML = methods;
}
</script>

```

```

</body>
</html>

```

- e. Write a program using string object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
    <title>String Object Example</title>
</head>
<body>

    <h2>String Object Properties and Methods</h2>

    <button onclick="showStringProperties()">Show String Properties</button>
    <button onclick="useStringMethods()">Use String Methods</button>

    <div id="output" style="margin-top: 20px; font-family: monospace;"></div>

    <script>
        // String property example
        function showStringProperties() {
            let str = "Hello, JavaScript!";
            let properties = `
                <strong>String:</strong> "${str}"<br>
                <strong>Length of string:</strong> ${str.length}<br>
            `;
            document.getElementById("output").innerHTML = properties;
        }

        // String methods examples
        function useStringMethods() {

```

```

let str = "Hello, JavaScript!";

let methods = `
  <strong>Original String:</strong> ${str}<br>
  <strong>toUpperCase():</strong> ${str.toUpperCase()}<br>
  <strong>toLowerCase():</strong> ${str.toLowerCase()}<br>
  <strong>charAt(7):</strong> ${str.charAt(7)}<br>
  <strong>indexOf('JavaScript'):</strong> ${str.indexOf('JavaScript')}<br>
  <strong>slice(7, 17):</strong> ${str.slice(7, 17)}<br>
  <strong>replace('JavaScript', 'JS'):</strong> ${str.replace('JavaScript', 'JS')}<br>
  <strong>split(', '):</strong> ${str.split(', ')}<br>
  <strong>includes('Hello'):</strong> ${str.includes('Hello')}<br>
`;

document.getElementById("output").innerHTML = methods;
}
</script>

</body>
</html>

```

- f. Write a program using regex object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
  <title>RegEx Object Example</title>
</head>
<body>

  <h2>RegEx Object Properties and Methods</h2>

  <button onclick="useRegex()">Test RegEx</button>

  <div id="output" style="margin-top: 20px; font-family: monospace;"></div>

  <script>
    function useRegex() {
      // Regular Expression to find "JavaScript" in a string
      let regex = /JavaScript/i; // 'i' => case-insensitive

      // Test string
      let str = "I am learning JavaScript programming.";

      // Properties
      let properties = `
        <strong>RegEx Pattern:</strong> ${regex}<br>

```

```

    <strong>Flags:</strong> ${regex.flags}<br>
    <strong>Source:</strong> ${regex.source}<br>
    `;

    // Methods
    let methods = `
        <strong>test():</strong> ${regex.test(str)}<br>
        <strong>exec():</strong> ${JSON.stringify(regex.exec(str))}<br>
    `;

    // test() → returns true if match found
    // exec() → returns array of match details or null

    document.getElementById("output").innerHTML = properties + methods;

    // Example of replace using regex
    let replaced = str.replace(regex, "JS");
    document.getElementById("output").innerHTML += `<strong>Replaced
String:</strong> ${replaced}`;
    }
</script>

</body>
</html>

```

- g. Write a program using date object properties and methods.

```

<!DOCTYPE html>
<html>
<head>
    <title>Date Object Example</title>
</head>
<body>

    <h2>Date Object Properties and Methods</h2>

    <button onclick="showDateProperties()">Show Current Date Properties</button>
    <button onclick="useDateMethods()">Use Date Methods</button>

    <div id="output" style="margin-top: 20px; font-family: monospace;"></div>

    <script>
        // Display date properties
        function showDateProperties() {
            let today = new Date();
            let properties = `
                <strong>Current Date & Time:</strong> ${today}<br>
            `;
        }
    </script>

```

```

        <strong>Timestamp (ms since Jan 1, 1970):</strong> ${today.getTime()}<br>
        <strong>Year:</strong> ${today.getFullYear()}<br>
        <strong>Month:</strong> ${today.getMonth() + 1} <em>(0 = January)</em><br>
        <strong>Date:</strong> ${today.getDate()}<br>
        <strong>Day:</strong> ${today.getDay()} <em>(0 = Sunday)</em><br>
        <strong>Hours:</strong> ${today.getHours()}<br>
        <strong>Minutes:</strong> ${today.getMinutes()}<br>
        <strong>Seconds:</strong> ${today.getSeconds()}<br>
    `;
    document.getElementById("output").innerHTML = properties;
}

// Demonstrate date methods
function useDateMethods() {
    let today = new Date();

    // Set a future date
    today.setFullYear(2030);
    today.setMonth(11); // December
    today.setDate(25);
    today.setHours(10);
    today.setMinutes(30);
    today.setSeconds(0);

    let methods = `
        <strong>Updated Future Date:</strong> ${today}<br>
        <strong>ISO String:</strong> ${today.toISOString()}<br>
        <strong>UTC String:</strong> ${today.toUTCString()}<br>
        <strong>Locale String:</strong> ${today.toLocaleString()}<br>
    `;

    document.getElementById("output").innerHTML = methods;
}
</script>

</body>
</html>

```

- h. Write a program to explain user-defined object by using properties, methods, accessors, constructors and display.

```

<!DOCTYPE html>
<html>
<head>
    <title>User-Defined Object Example</title>

```

```
</head>
```

```
<body>
```

```
<h2>User-Defined Object with Properties, Methods, Accessors, and Constructor</h2>
```

```
<button onclick="createUser()">Create & Display User</button>
```

```
<div id="output" style="margin-top: 20px; font-family: monospace;"></div>
```

```
<script>
```

```
    // Constructor function to create a User object
```

```
    function User(name, age, email) {
```

```
        // Properties
```

```
        this.name = name;
```

```
        this.age = age;
```

```
        this.email = email;
```

```
        // Method
```

```
        this.greet = function() {
```

```
            return `Hello, my name is ${this.name}`;
```

```
        };
```

```
        // Accessors (getter & setter)
```

```
        this.getAge = function() {
```

```
            return this.age;
```

```
        };
```

```
        this.setAge = function(newAge) {
```

```
            if(newAge > 0) {
```

```
                this.age = newAge;
```

```
            } else {
```

```
                alert("Invalid age!");
```

```
            }
```

```
        };
```

```
        // Display Method
```

```
        this.displayInfo = function() {
```

```
            return `
```

```
                Name: ${this.name}<br>
```

```
                Age: ${this.age}<br>
```

```
                Email: ${this.email}<br>
```

```
                Greeting: ${this.greet()}`
```

```
            `;
```

```
        };
```

```
    }
```

```

// Function to create a user and display
function createUser() {
    // Create a new User object
    let user1 = new User("Alice", 25, "alice@example.com");

    // Update age using setter
    user1.setAge(26);

    // Display user info
    document.getElementById("output").innerHTML = user1.displayInfo();
}
</script>

</body>
</html>

```

8. JavaScript Conditional Statements and Loops

- Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words “LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”.

```

<!DOCTYPE html>
<html>
<head>
    <title>Find Largest Number</title>
    <script>
        function findLargestNumber() {
            // Get values from input boxes
            var num1 = parseInt(document.getElementById("num1").value);
            var num2 = parseInt(document.getElementById("num2").value);
            var num3 = parseInt(document.getElementById("num3").value);

            // Compare the numbers
            if (num1 === num2 && num2 === num3) {
                alert("EQUAL NUMBERS");
            } else {
                var largest = num1;
                if (num2 > largest) {
                    largest = num2;
                }
                if (num3 > largest) {
                    largest = num3;
                }
                alert(largest + " LARGER NUMBER");
            }
        }
    </script>

```



```

    }
</script>
</head>
<body>
  <h2>Enter Three Integers</h2>
  <form onsubmit="findLargestNumber(); return false;">
    <label>Enter first number:</label>
    <input type="number" id="num1" required><br><br>

    <label>Enter second number:</label>
    <input type="number" id="num2" required><br><br>

    <label>Enter third number:</label>
    <input type="number" id="num3" required><br><br>

    <input type="submit" value="Find Larger Number">
  </form>
</body>
</html>

```

b. Write a program to display week days using switch case.

```

<!DOCTYPE html>
<html>
<head>
  <title>Weekday Display</title>
  <script>
    function displayWeekday() {
      var dayNumber = parseInt(document.getElementById("day").value);
      var message = "";

      switch(dayNumber) {
        case 1:
          message = "Sunday";
          break;
        case 2:
          message = "Monday";
          break;
        case 3:
          message = "Tuesday";
          break;
        case 4:
          message = "Wednesday";
          break;
        case 5:

```

```

        message = "Thursday";
        break;
    case 6:
        message = "Friday";
        break;
    case 7:
        message = "Saturday";
        break;
    default:
        message = "Invalid input! Enter a number between 1 and 7.";
    }

    alert(message); // Display the result in a dialog box
}
</script>
</head>
<body>
    <h2>Enter a Number (1 to 7) to Display the Weekday</h2>
    <form onsubmit="displayWeekday(); return false;">
        <label>Day Number (1-7):</label>
        <input type="number" id="day" required><br><br>
        <input type="submit" value="Show Weekday">
    </form>
</body>
</html>

```

- c. Write a program to print 1 to 10 numbers using for, while and do-while loops.

```

<!DOCTYPE html>
<html>
<head>
    <title>Print 1 to 10 using Loops</title>
</head>
<script>
    function printNumbers() {
        let output = "";

        // Using for loop
        output += "<strong>Using for loop:</strong><br>";
        for (let i = 1; i <= 10; i++) {
            output += i + " ";
        }
        output += "<br><br>";

        // Using while loop
        output += "<strong>Using while loop:</strong><br>";
        let j = 1;
        while (j <= 10) {

```

```

        output += j + " ";
        j++;
    }
    output += "<br><br>";

    // Using do-while loop
    output += "<strong>Using do-while loop:</strong><br>";
    let k = 1;
    do {
        output += k + " ";
        k++;
    } while (k <= 10);

    // Display the output on the webpage
    document.getElementById("result").innerHTML = output;
}
</script>
</head>
<body>
    <h2>Print Numbers from 1 to 10 using Loops</h2>
    <button onclick="printNumbers()">Click to Show Output</button>
    <div id="result" style="margin-top:20px; font-size:18px;"></div>
</body>
</html>

```

- d. Write program to print data in object using for-in, for-each and for-of loops

```

<!DOCTYPE html>
<html>
<head>
    <title>Object Data Looping</title>
</head>
<script>
    function displayObjectData() {
        // Sample object
        const student = {
            name: "Alice",
            age: 20,
            course: "Computer Science",
            city: "Chennai"
        };

        let output = "";

        // Using for-in loop (best for objects)
        output += "<strong>Using for-in loop:</strong><br>";
        for (let key in student) {
            output += key + ": " + student[key] + "<br>";
        }
    }

```

```

    }
    output += "<br>";

    // Convert object values to an array for forEach and for-of
    const values = Object.values(student);

    // Using forEach loop (used with arrays)
    output += "<strong>Using forEach loop:</strong><br>";
    values.forEach(function(value) {
        output += value + "<br>";
    });
    output += "<br>";

    // Using for-of loop (used with arrays)
    output += "<strong>Using for-of loop:</strong><br>";
    for (let value of values) {
        output += value + "<br>";
    }

    // Display the output
    document.getElementById("result").innerHTML = output;
}
</script>
</head>
<body>
    <h2>Display Object Data Using Different Loops</h2>
    <button onclick="displayObjectData()">Show Object Data</button>
    <div id="result" style="margin-top: 20px; font-size: 16px;"></div>
</body>
</html>

```

- e. Develop a program to determine whether a given number is an ‘ARMSTRONG NUMBER’ or not. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e., $1^3 + 5^3 + 3^3 = 153$]

```

<!DOCTYPE html>
<html>
<head>
    <title>Armstrong Number Checker</title>
    <script>
        function checkArmstrong() {
            let num = parseInt(document.getElementById("number").value);
            let originalNum = num;
            let sum = 0;

            let digits = num.toString().length; // To handle n-digit numbers (like 9474, 4-digit)

```

```

    while (num > 0) {
        let digit = num % 10;
        sum += Math.pow(digit, digits); // cube if 3 digits, power of n otherwise
        num = Math.floor(num / 10);
    }

    if (sum === originalNum) {
        alert(originalNum + " is an ARMSTRONG NUMBER");
    } else {
        alert(originalNum + " is NOT an Armstrong Number");
    }
}
</script>
</head>
<body>
    <h2>Check Armstrong Number</h2>
    <form onsubmit="checkArmstrong(); return false;">
        <label>Enter a number:</label>
        <input type="number" id="number" required><br><br>
        <input type="submit" value="Check">
    </form>
</body>
</html>

```

- f. Write a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1-100's, 1-50's, 1- 10's, 1-2's & 1-1's)

```

function displayDenominations(amount) {
    const denominations = [100, 50, 20, 10, 5, 2, 1];
    const result = {};

    for (let i = 0; i < denominations.length; i++) {
        let note = denominations[i];
        if (amount >= note) {
            result[note] = Math.floor(amount / note);
            amount = amount % note;
        }
    }

    // Display the result
    console.log("Denominations:");
    for (let note in result) {
        console.log(`${result[note]} - ₹${note}'s`);
    }
}

```

```
}
```

```
// Example usage:
```

```
let depositedAmount = 163;
```

```
displayDenominations(depositedAmount);
```

9. Javascript Functions and Events

a. Design a appropriate function should be called to display

- i. Factorial of that number
- ii. Fibonacci series up to that number
- iii. Prime numbers up to that number
- iv. Is it palindrome or not

```
// i. Function to find factorial
```

```
function factorial(n) {  
  if (n < 0) return "Invalid input";  
  let fact = 1;  
  for (let i = 1; i <= n; i++) {  
    fact *= i;  
  }  
  return fact;  
}
```

```
// ii. Function to print Fibonacci series up to n
```

```
function fibonacci(n) {  
  let fib = [0, 1];  
  let result = [];  
  
  for (let i = 0; i < n; i++) {  
    if (i <= 1) {  
      result.push(fib[i]);  
    } else {  
      fib[i] = fib[i - 1] + fib[i - 2];  
      if (fib[i] <= n) {  
        result.push(fib[i]);  
      } else {  
        break;  
      }  
    }  
  }  
  
  return result;  
}
```

```
// iii. Function to find prime numbers up to n
```

```

function primesUpTo(n) {
    let primes = [];

    for (let i = 2; i <= n; i++) {
        let isPrime = true;
        for (let j = 2; j <= Math.sqrt(i); j++) {
            if (i % j === 0) {
                isPrime = false;
                break;
            }
        }
        if (isPrime) {
            primes.push(i);
        }
    }

    return primes;
}

// iv. Function to check palindrome
function isPalindrome(n) {
    let original = n.toString();
    let reversed = original.split("").reverse().join("");
    return original === reversed;
}

// === Main Call ===
let num = 121; // You can change this number to test with different input

console.log("Factorial of", num, "is:", factorial(num));
console.log("Fibonacci series up to", num, "is:", fibonacci(num));
console.log("Prime numbers up to", num, "are:", primesUpTo(num));
console.log(num, "is palindrome?", isPalindrome(num));

```

- b. Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display
- Factorial of that number
 - Fibonacci series up to that number
 - Prime numbers up to that number
 - Is it palindrome or not

```

<!DOCTYPE html>
<html>
<head>
    <title>Number Operations</title>
    <style>

```

```

    body {
        font-family: Arial, sans-serif;
        padding: 20px;
    }
    input, button {
        padding: 10px;
        margin: 5px;
    }
    #result {
        margin-top: 15px;
        font-weight: bold;
    }
</style>
</head>
<body>

    <h2>Enter a Number:</h2>
    <input type="number" id="numberInput" placeholder="Enter number" />

    <br/>

    <button onclick="showFactorial()">Factorial</button>
    <button onclick="showFibonacci()">Fibonacci</button>
    <button onclick="showPrimes()">Prime</button>
    <button onclick="checkPalindrome()">Palindrome</button>

    <div id="result"></div>

    <script>
        function showFactorial() {
            let n = parseInt(document.getElementById("numberInput").value);
            let result = 1;
            for (let i = 1; i <= n; i++) {
                result *= i;
            }
            document.getElementById("result").innerText = `Factorial of ${n} is:
    ${result}`;
        }

        function showFibonacci() {
            let n = parseInt(document.getElementById("numberInput").value);
            let a = 0, b = 1, fib = [];
            while (a <= n) {
                fib.push(a);
                [a, b] = [b, a + b];
            }

```



```

        document.getElementById("result").innerText = `Fibonacci series up to ${n}
is: ${fib.join(', ')}`;
    }

```

```

    function showPrimes() {
        let n = parseInt(document.getElementById("numberInput").value);
        let primes = [];
        for (let i = 2; i <= n; i++) {
            let isPrime = true;
            for (let j = 2; j <= Math.sqrt(i); j++) {
                if (i % j === 0) {
                    isPrime = false;
                    break;
                }
            }
            if (isPrime) {
                primes.push(i);
            }
        }
        document.getElementById("result").innerText = `Prime numbers up to ${n}
are: ${primes.join(', ')}`;
    }

```

```

    function checkPalindrome() {
        let n = document.getElementById("numberInput").value;
        let reversed = n.split("").reverse().join("");
        let isPalindrome = n === reversed;
        document.getElementById("result").innerText = `${n} is ${isPalindrome ? " :
'not ' } a Palindrome.`;
    }
</script>

```

```

</body>
</html>

```

- c. Write a program to validate the following fields in a registration page
- Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
 - Mobile (only numbers and length 10 digits)
 - E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

```

<!DOCTYPE html>
<html>
<head>
    <title>Registration Form Validation</title>
    <style>

```

```

body {
    font-family: Arial;
    padding: 20px;
}
input, button {
    padding: 10px;
    margin: 5px 0;
    width: 300px;
}
.error {
    color: red;
    font-size: 14px;
}
.success {
    color: green;
}
</style>
</head>
<body>

<h2>Registration Form</h2>

<form onsubmit="return validateForm()">
    <label>Name:</label><br>
    <input type="text" id="name"><br>
    <span class="error" id="nameError"></span><br>

    <label>Mobile:</label><br>
    <input type="text" id="mobile"><br>
    <span class="error" id="mobileError"></span><br>

    <label>Email:</label><br>
    <input type="text" id="email"><br>
    <span class="error" id="emailError"></span><br>

    <button type="submit">Register</button>
</form>

<div id="result" class="success"></div>

<script>
    function validateForm() {
        let isValid = true;

        // Clear previous messages
        document.getElementById("nameError").innerText = "";

```

```

document.getElementById("mobileError").innerText = "";
document.getElementById("emailError").innerText = "";
document.getElementById("result").innerText = "";

// Name validation
let name = document.getElementById("name").value.trim();
let nameRegex = /^[A-Za-z][A-Za-z0-9]{5,}$/;
if (!nameRegex.test(name)) {
    document.getElementById("nameError").innerText = "Invalid name (must
start with a letter and be at least 6 characters)";
    isValid = false;
}

// Mobile validation
let mobile = document.getElementById("mobile").value.trim();
let mobileRegex = /^[0-9]{10}$/;
if (!mobileRegex.test(mobile)) {
    document.getElementById("mobileError").innerText = "Mobile number
must be 10 digits only";
    isValid = false;
}

// Email validation
let email = document.getElementById("email").value.trim();
let emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-z]{2,4}$/;
if (!emailRegex.test(email)) {
    document.getElementById("emailError").innerText = "Invalid email
format";
    isValid = false;
}

if (isValid) {
    document.getElementById("result").innerText = "Registration Successful!";
}

return isValid;
}
</script>

</body>
</html>

```